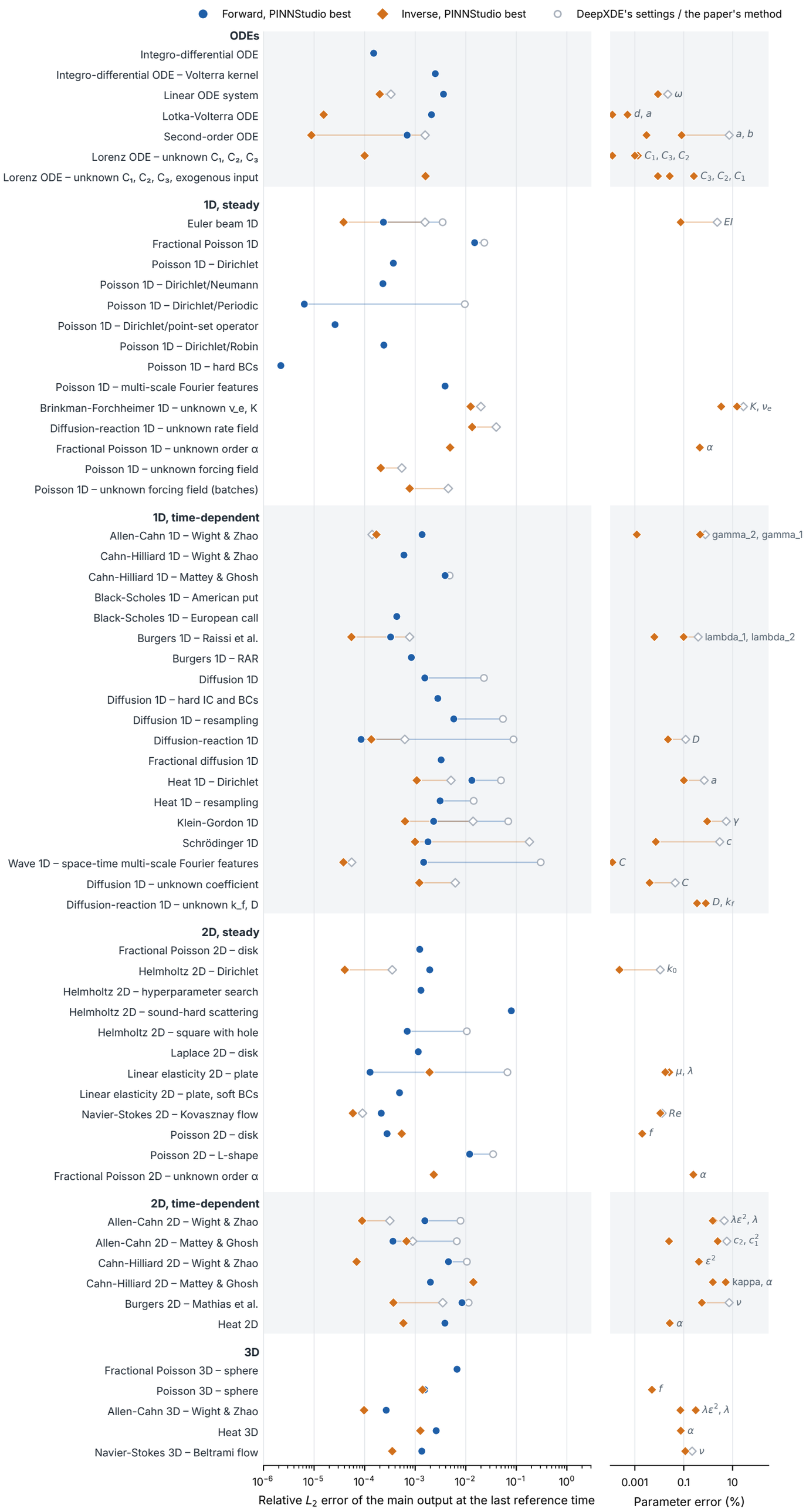

Forward, PINNStudio best
Inverse, PINNStudio best
DeepXDE's settings / the paper's method
ODEs
Integro-differential ODE
Integro-differential ODE – Volterra kernel
Linear ODE system
Lotka-Volterra ODE
Second-order ODE
Lorenz ODE – unknown C_1, C_2, C_3
Lorenz ODE – unknown C_1, C_2, C_3, exogenous input
1D, steady
Euler beam 1D
Fractional Poisson 1D
Poisson 1D – Dirichlet
Poisson 1D – Dirichlet/Neumann
Poisson 1D – Dirichlet/Periodic
Poisson 1D – Dirichlet/point-set operator
Poisson 1D – Dirichlet/Robin
Poisson 1D – hard BCs
Poisson 1D – multi-scale Fourier features
Brinkman-Forchheimer 1D – unknown v_e, K
Diffusion-reaction 1D – unknown rate field
Fractional Poisson 1D – unknown order α
Poisson 1D – unknown forcing field
Poisson 1D – unknown forcing field (batches)
1D, time-dependent
Allen-Cahn 1D – Wight & Zhao
Cahn-Hilliard 1D – Wight & Zhao
Cahn-Hilliard 1D – Mattey & Ghosh
Black-Scholes 1D – American put
Black-Scholes 1D – European call
Burgers 1D – Raissi et al.
Burgers 1D – RAR
Diffusion 1D
Diffusion 1D – hard IC and BCs
Diffusion 1D – resampling
Diffusion-reaction 1D
Fractional diffusion 1D
Heat 1D – Dirichlet
Heat 1D – resampling
Klein-Gordon 1D
Schrödinger 1D
Wave 1D – space-time multi-scale Fourier features
Diffusion 1D – unknown coefficient
Diffusion-reaction 1D – unknown k_f, D
2D, steady
Fractional Poisson 2D – disk
Helmholtz 2D – Dirichlet
Helmholtz 2D – hyperparameter search
Helmholtz 2D – sound-hard scattering
Helmholtz 2D – square with hole
Laplace 2D – disk
Linear elasticity 2D – plate
Linear elasticity 2D – plate, soft BCs
Navier-Stokes 2D – Kovasznay flow
Poisson 2D – disk
Poisson 2D – L-shape
Fractional Poisson 2D – unknown order α
2D, time-dependent
Allen-Cahn 2D – Wight & Zhao
Allen-Cahn 2D – Mattey & Ghosh
Cahn-Hilliard 2D – Wight & Zhao
Cahn-Hilliard 2D – Mattey & Ghosh
Burgers 2D – Mathias et al.
Heat 2D
3D
Fractional Poisson 3D – sphere
Poisson 3D – sphere
Allen-Cahn 3D – Wight & Zhao
Heat 3D
Navier-Stokes 3D – Beltrami flow
ω
d, a
a, b
C_1, C_3, C_2
C_3, C_2, C_1
EI
K, ν_e
α
gamma_2, gamma_1
lambda_1, lambda_2
D
a
γ
c
C
C
D, k_f
k_0
μ, λ
Re
f
α
$\lambda\varepsilon^2$, λ
c_2, c_1^2
ε^2
kappa, α
ν
α
f
$\lambda\varepsilon^2$, λ
α
ν
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Relative L_2 error of the main output at the last reference time
0.001
0.1
10
Parameter error (%)